CHANDIGARH UNIVERSITY

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Subject: Technical Training

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Q1 As a DBA in a finance enterprise, compare Role based access control with user based privilege assignment which scales better and why?

Ans Roles Based Access control (RBAC):-
Privileges are assigned to Roles (for eg: Auditor, DB admin) and users are assigned those roles.

→ Highly Scalable: one role can be assigned to hundreds or thousands of users.

→ easy maintainance: change a role once, it updates access role for all users

→ stronger security: supports least privilege and separation of duties

→ Audit-friendly: easier to review who has what access (critical in finance)

→ cons: Initial role design takes time for planning

User based Privilege Assignment:-
Privileges are granted directly to individual users.

→ Pros: 1. simple for very small system.
2. Fine grained, user specific control

Cons: 1. Poor scalability; every user needs manual privilege setup
2. Hard to manage - changes must be repeated for each user
3. error-prone → high risk of privilege creep
4. Difficult Auditing → messy in compliance-heavy environments.

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example: for each user.
create user 'alice' identified by 'pwd1';
create user 'hobby' identified by 'pwd b';
grant select, insert on accounts to alice;
grant select on accounts to hobby;

→ Hence Role based access control scales better because of following reasons
1. Financial enterprises have large dynamics user basis
2. RBAC induces administratives overhead
3. supports regulatary compliance.
4. makes onboarding, Role changes and off boarding fast and safe.

example of RBAC
create ROLE teller_role;
grant select, insert on accounts to teller role,

create user 'alice' identified by 'pwd1'
grant teller role to alice;

Q2 Design a three tier architecture for a wealth management system explain security enforcement at each tier

Ans Three tier Architecture consist of
1. client → ② server → ① Database.

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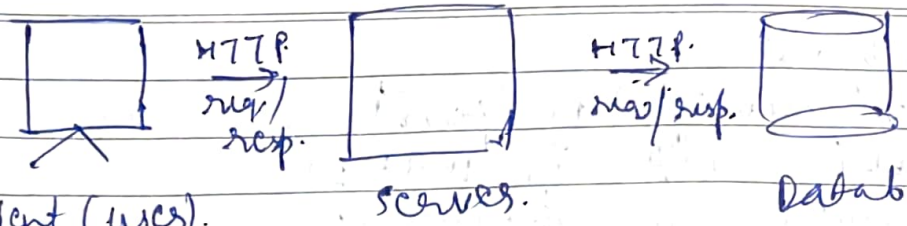

A three tier architecture separates the system into presentation, application and database tiers to improve security, scalability and maintainability.

1. Presentation tier (client layer).
   ROLE:- 1. user interface for clients, relationship mangers and administrators.
   2. web browsers, mobile app dashboards for portfolio transactions reports.
   3. communicate with the application tier via secure API's
   4. This tier provides the user interface for clients administrators and etc. to interact with the system.

Security enforcements:
- user administration - (login, MFA, OTP, biometrics)
- Input validation to prevent malicious data entry
- HTTPS/TLS encryption for secure communication
- session management with time outs and secure cookies
- client side access control (UI restrictions based on Roles)

2. Application tier (Business logic layer):-
   This tier contains the core business logic of the

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wealth management system. It process requests from the presentation tier and enforces financial rules.

1. Portfolio calculations and risk analysis
2. Transaction validation and compliance checks
3. Role based authorised decisions
4. Interactions with database tier

Security:-

1. Role based access control for clients, advisors.
2. Authorization checks on request
3. Protection against (SQL injection, DDOS)

3. Database tier (Data layer):-

This tier stores and manages all sensitive financial data such as client profiles, portfolio transaction and compliance logs.

1. Persistent storage of structured financial Data
2. Data Retrival and updates
3. Backup & Recovery

Security:-

1. Strong access controls (Roles, schemas least privilege)
2. encryption at rest for sensitive data
3. Database authentication and authorisation
4. Stored procedures views to restrict direct table access
5. Regular backups and integrity checks

Q3 explain the ANSI-SPARC 3 level architecture with a diagram. How does it help in achieving logical and physical data independence.

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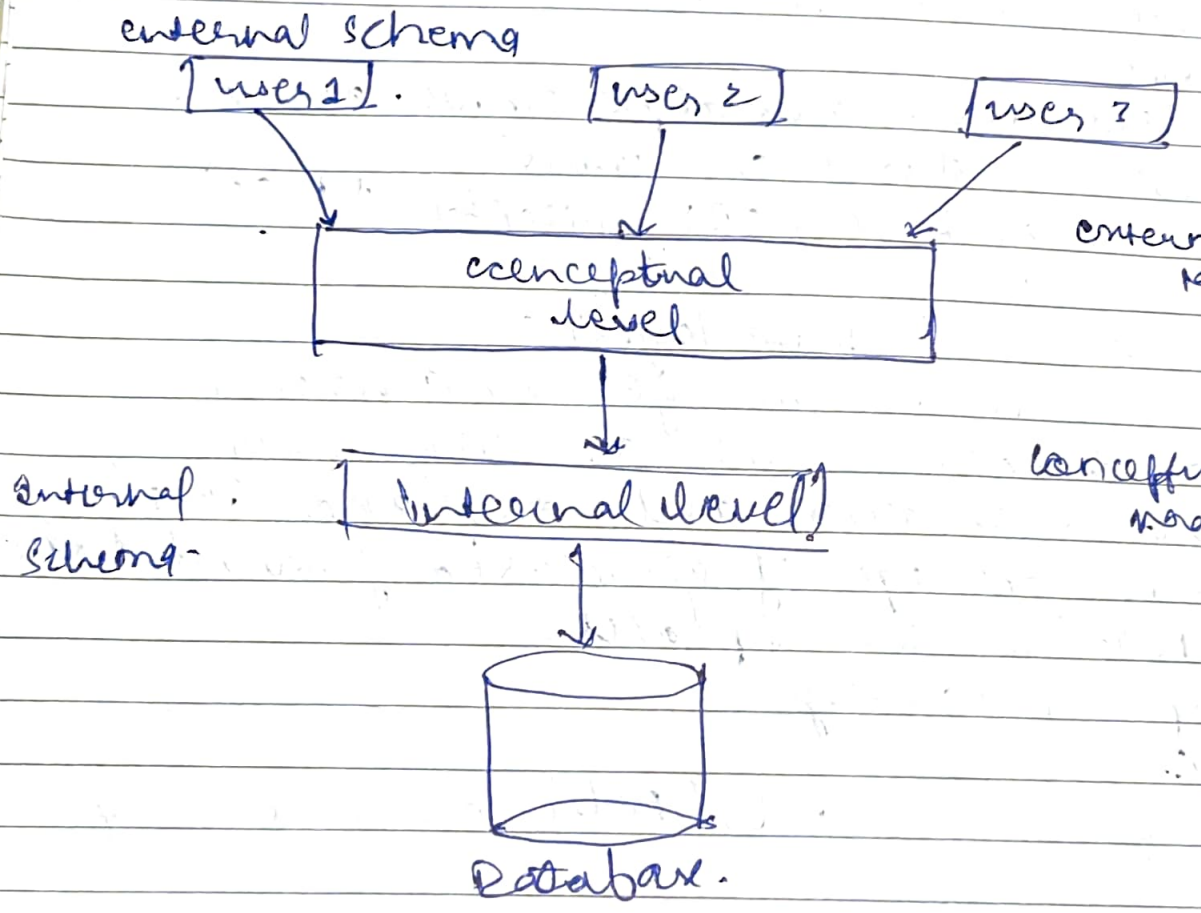
Ans ANSI-SPARC 3 level architecture is a database design framework that separates a DBS into 3 abstraction levels. The goal is to hide complexity improve security and achieve data independence.



1. external level (views level):-
The external level represents the user's view of database and defines how individual users or group of users view the data each external schemes consists of only the data relevant to particular user, hiding the rest of the DB for simplicity and security, multiple external schemes can exist for same DB allowing

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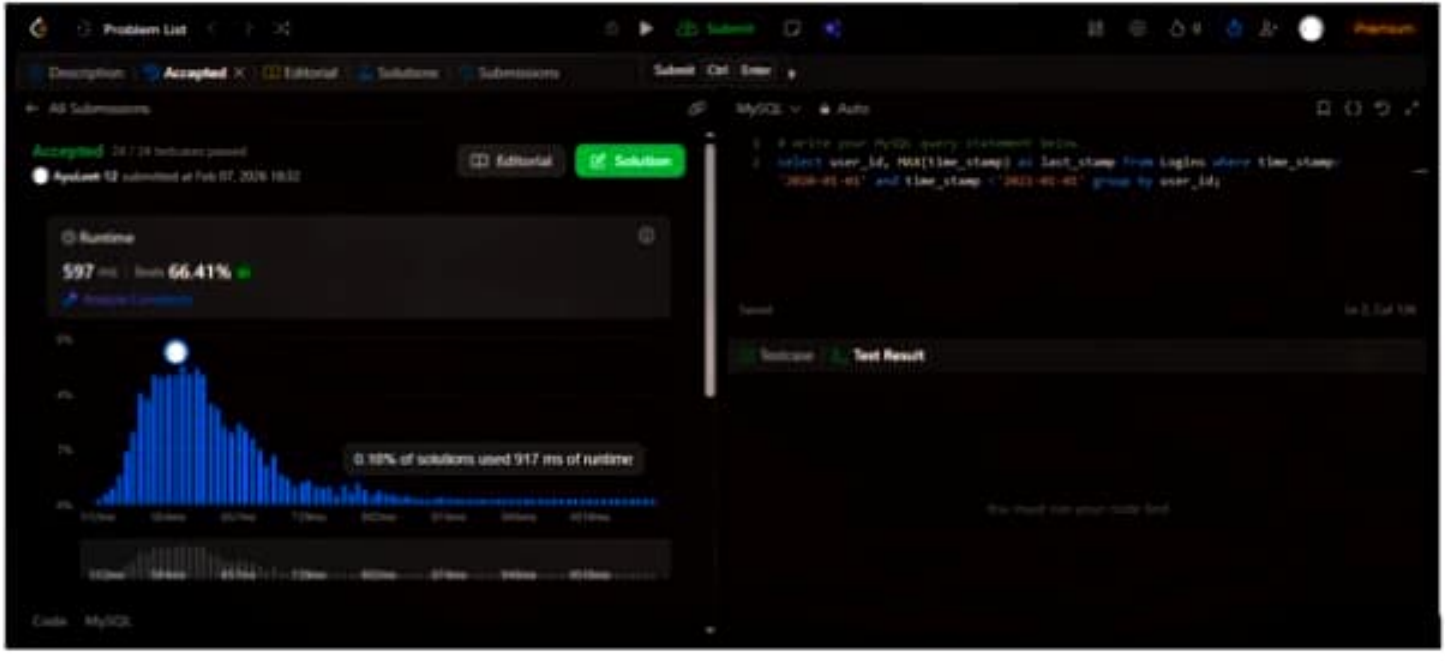
different users to have different customized views. This level helps in enforcing access control and provide data abstractions by shielding users from the complexity of the complete DB structure.

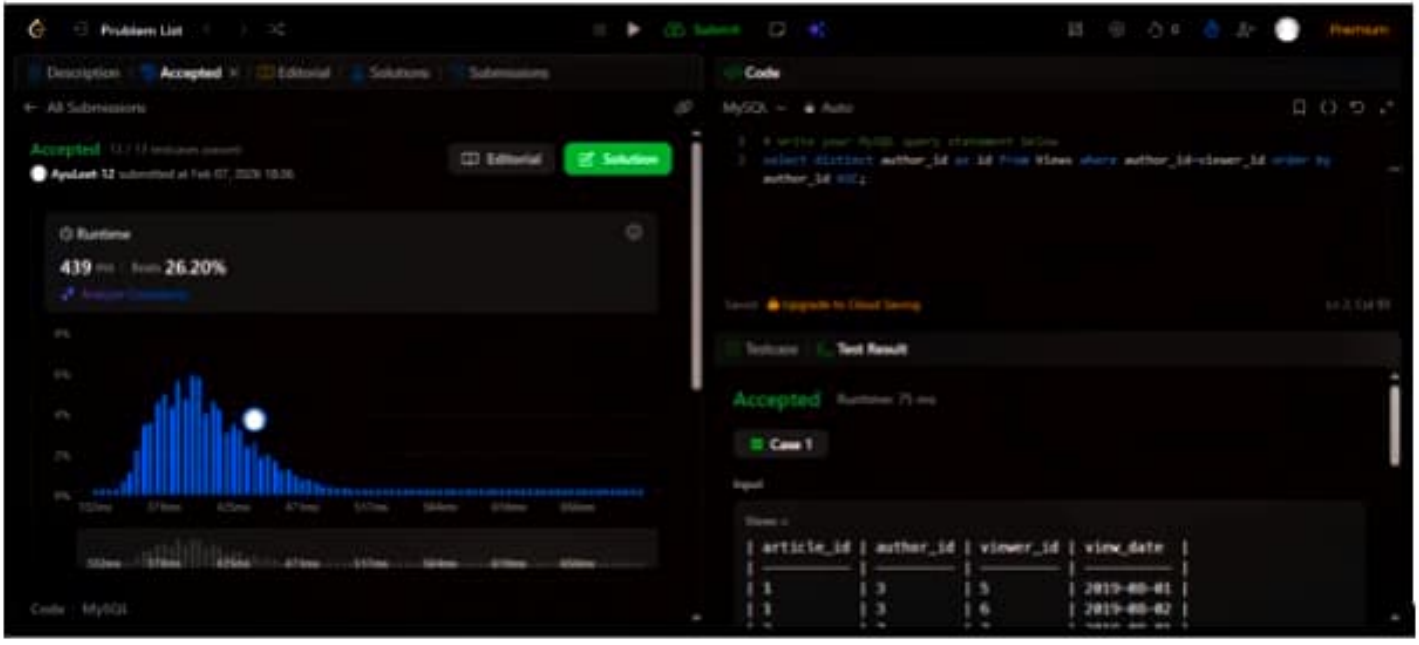
2. Conceptual level :- (logical level)
This level describes the structure of the entire DB as seen by DBA. It defines what data is stored, the relationship among data, the constraints applied without considering how the data is stored. There is only one conceptual schema per the DB and it serves the bridge between user views and physical storage. This level ensures data consistency by maintaining complete logical view of database.

3. Internal level (physical level)
This level specifies how the data is physically stored in the DBS. It deals with low level storage details such as file organisation, indexing methods, access paths and data compression. This level is concerned with performance storage optimization. Changes made at internal level does not affect the logical structure of user views, thereby supporting physical data independence.

Q4) Leetcode The latest login 2020

select user_id, MAX(time_stamp) as last_stamp from Logins where time_stamp> '2020-01-01' and time_stamp <'2021-01-01' group by user_id;



Q5) Leetcode - Article Views 1

select distinct author_id as id from Views where author_id=viewer_id order by author_id ASC;